

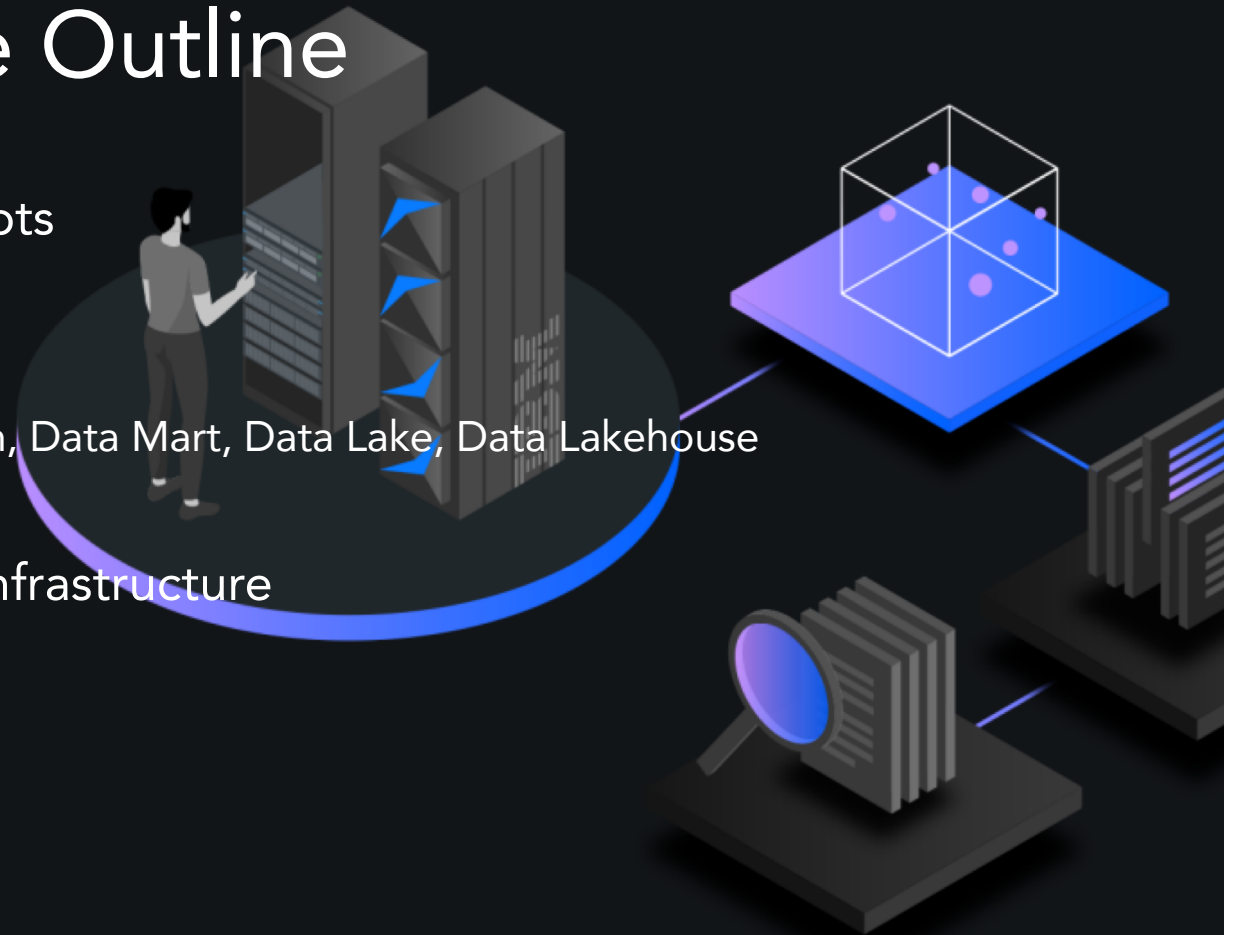


Redshift

Datawarehouse on AWS Cloud

Redshift Course Outline

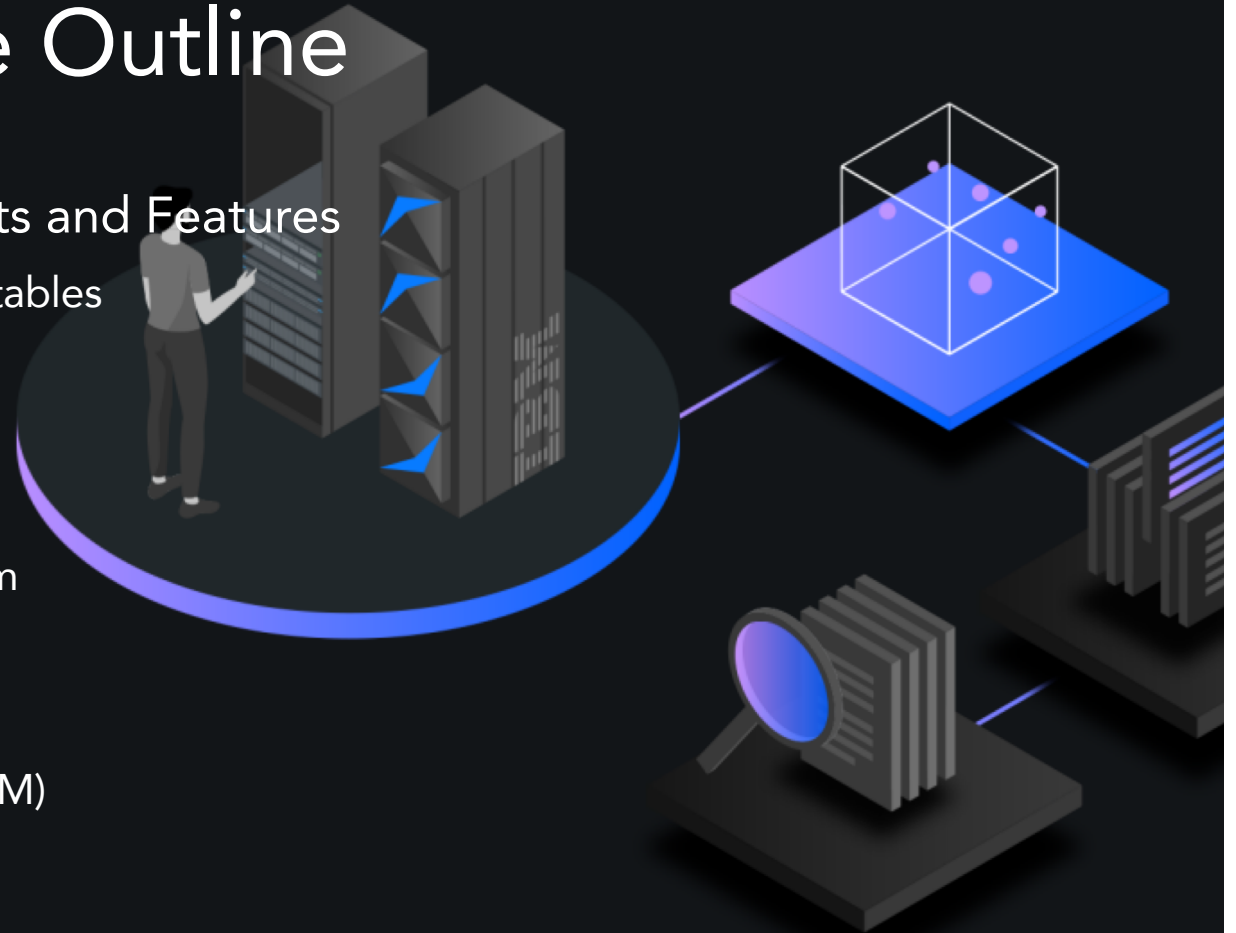
- Section 1 – General concepts
 - Source systems
 - OLTP, OLAP
 - Data Warehouse, Data Mesh, Data Mart, Data Lake, Data Lakehouse
 - Data Engineer
- Section 2 – Redshift & its infrastructure
 - Architecture
 - Nodes, VPC, MPP
 - Types of clusters
 - Resizing
 - Free tier



Redshift Course Outline

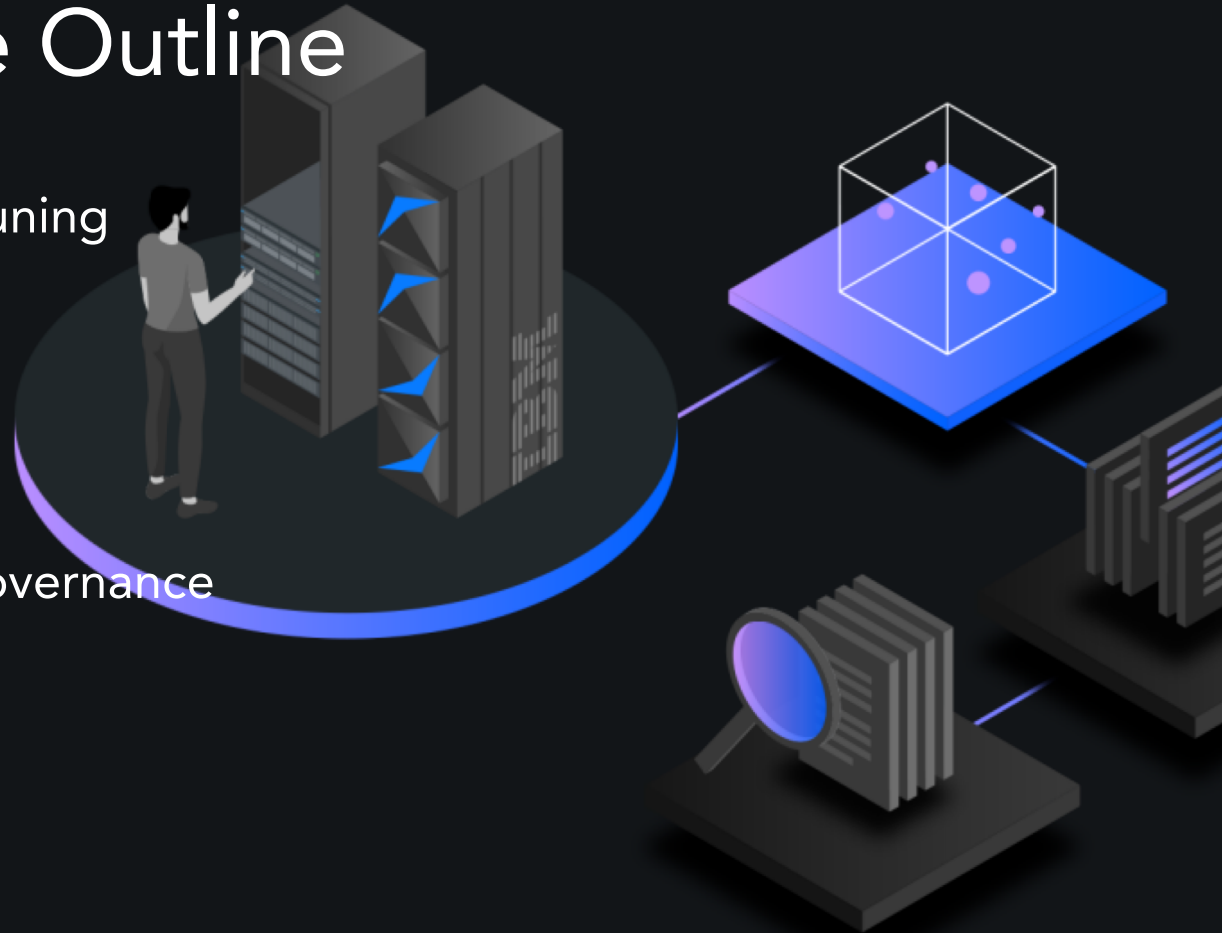
- Section 3 – Redshift Objects and Features

- Create objects – Database, tables
- Load/Unload data
- Views, Materialized Views
- Streaming ingestion
- Federated queries, Spectrum
- Data sharing
- Lakehouse with Redshift
- Workload Management (WLM)



Redshift Course Outline

- Section 4 – Performance Tuning
 - EXPLAIN
 - Table optimization
 - Maintenance operations
 - Query tuning
- Section 5 – Security and governance
 - Users, groups, roles
 - RLS, CLS, Dynamic masking
 - AWS Lake Formation
 - AWS Glue Catalog



Redshift Course Outline

- Section 6 – Redshift Serverless
- Section 7 – Integration
 - Spark
 - EventBridge
 - Iceberg (open table format)
 - Zero-ETL
 - QuickSight
- Section 8 – Operations
 - Monitoring
 - Pricing



Redshift Course Outline

- Section 9 – Additional Information

- Best practices
- Redshift limits
- User defined functions
- Stored procedures
- Spatial data
- Reserved nodes
- Comparison with Snowflake

- Section 10 – New Features

- Continuous update...

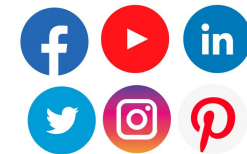




Section 1 – General Concepts

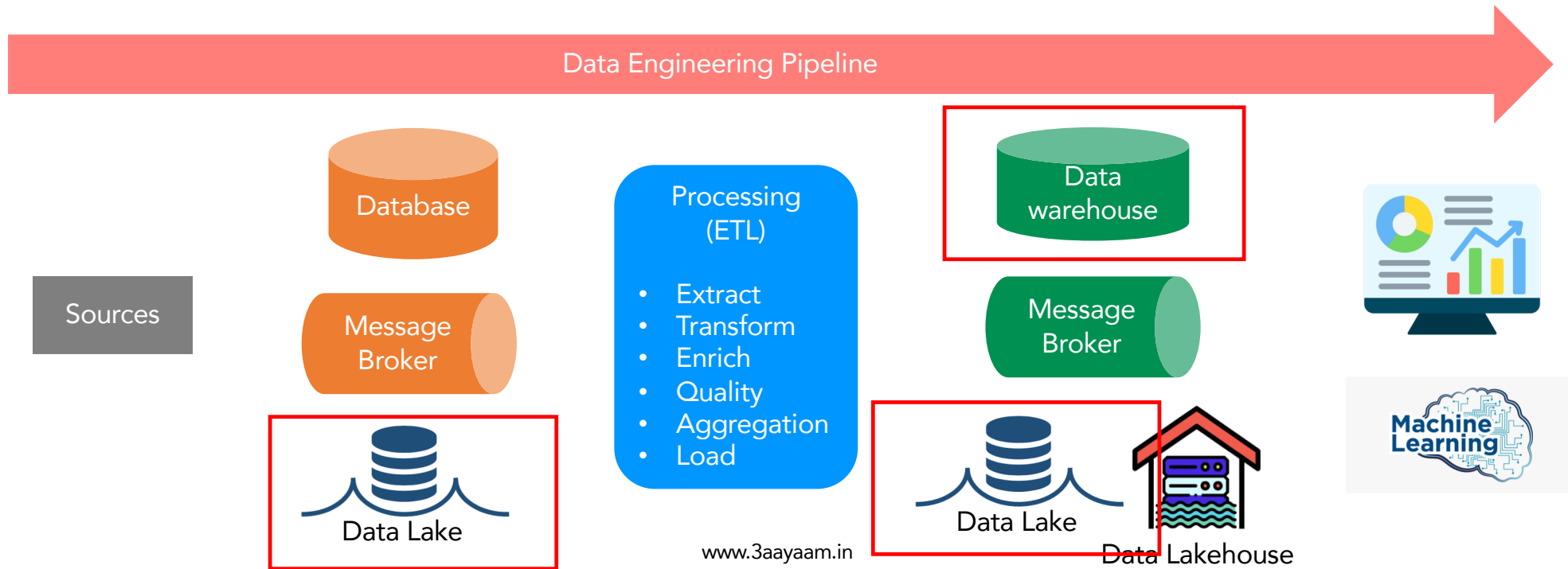
Genesis of Data

- Mobile applications
- Web applications
- Smart devices (TV, Watch, Fridge, Wearables)
- IoT devices (sensors)
- EVs
- Social media



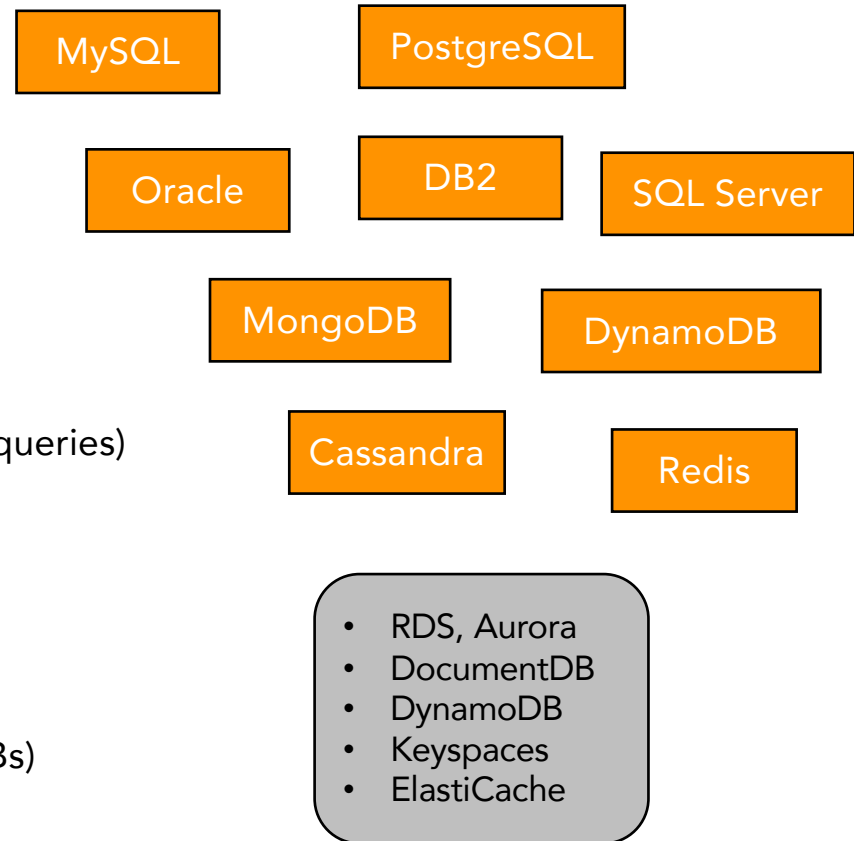
Data Lifecycle

- Generate
- Collect
- Process
- Store
- Analyze
- Consume



Operational Databases (OLTP)

- Operational datastores
- Runs the business
- Day-to-day processing
- Process database transactions
- Data comes in
- Current data
- High concurrency (thousands to millions of users and queries)
- Simple queries
- 1s - 10s - 100s output rows
- Small operations (read and write)
- Millisecond processing time
- Hundreds of Gigabytes (GBs) or Units of Terabytes (TBs)

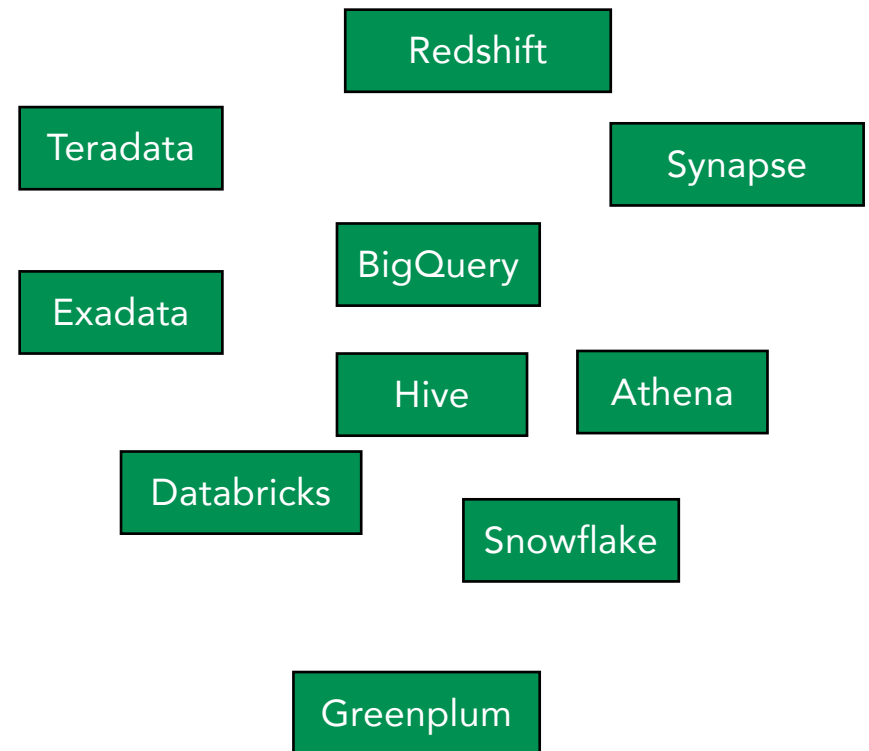


Analytical Databases (OLAP)

- Historical processing
- Analyze the business
- Historical data
- Data goes out
- Lower concurrency (tens to few thousands)
- Complex queries
- Tens of thousands of output rows
- Mostly read operations
- Processing time – seconds to hours
- 10s of Terabytes (TB) to Petabytes (PB)

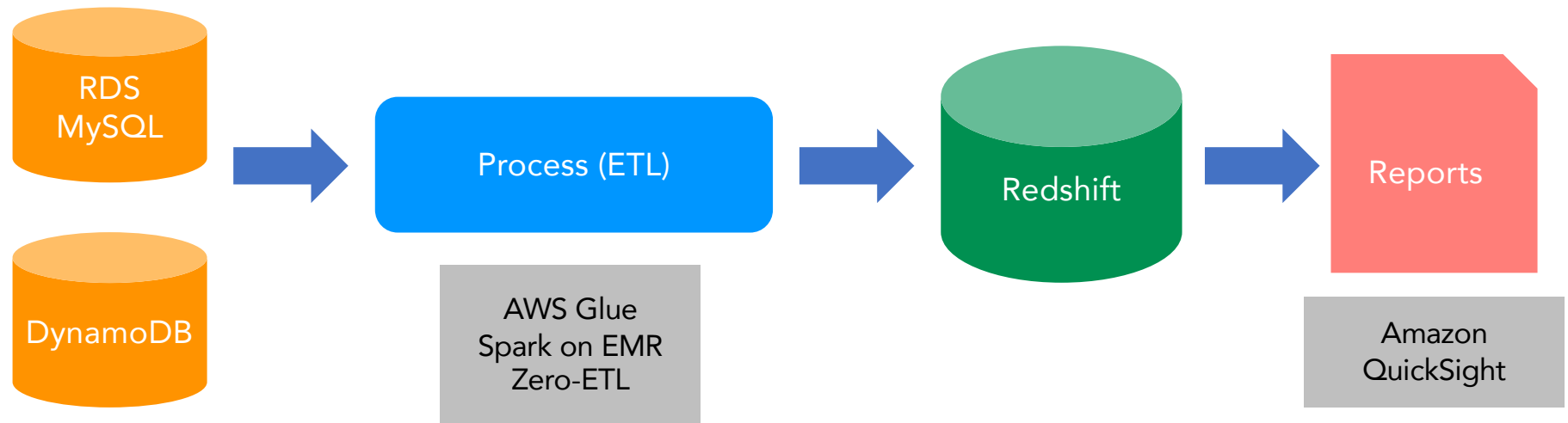
Analytical Databases (OLAP)

- Marketing, sales, campaign
- Performance of business
- Combine data from diverse sources
- Learn about business
- Gain intelligence on business activities
- Analytics & Reporting
- **Data Warehouse**



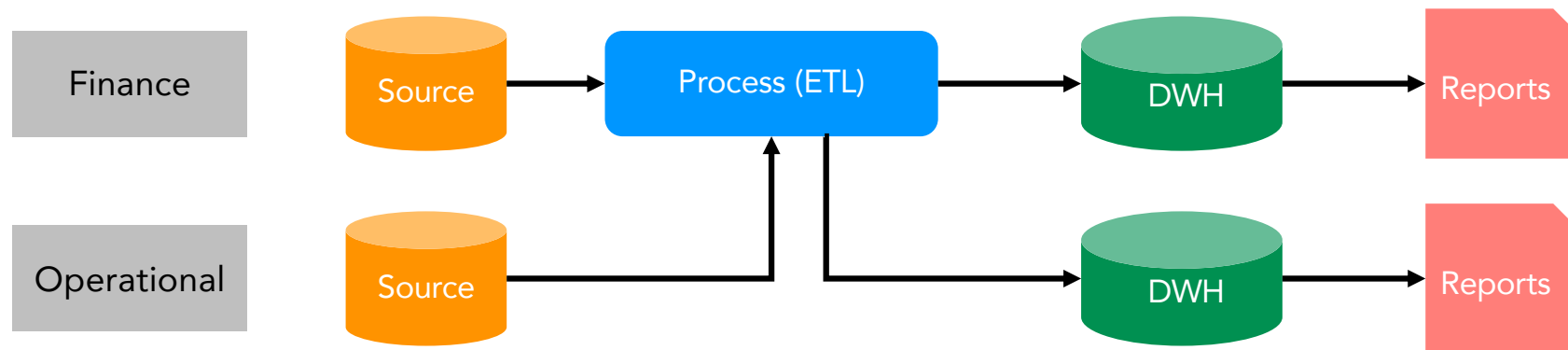
OLTP to OLAP

- ETL (Extract, Transform and Load)
- Processing (transformation)



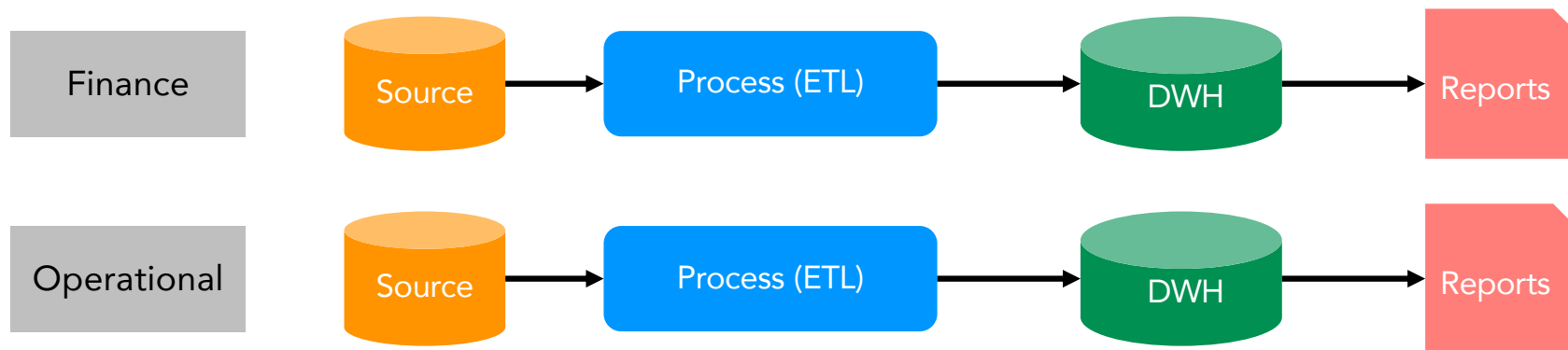
Data Mart, Data Mesh

A data mart is a subset of a data warehouse focused on a particular line of business, department or subject area.



Different data warehouse for domain, line of business or subject area etc.

Data Mart, Data Mesh



A data mesh is a decentralized data management, which means each line of business will have their entire data pipeline that is completely different than the other business domains.

Data Lake

- A data lake is a centralized repository that stores structured, semi-structured and unstructured data at any scale.
- CSV, XLS, Parquet, Avro, ORC
- JSON, XML
- JPG, PNG, DOCX, MP4, PDF, Binary data
- Raw data
- Processed data
- Read directly from data lake using
 - Native tools (S3 SELECT)
 - Presto
 - Athena
 - Redshift
 - BigQuery

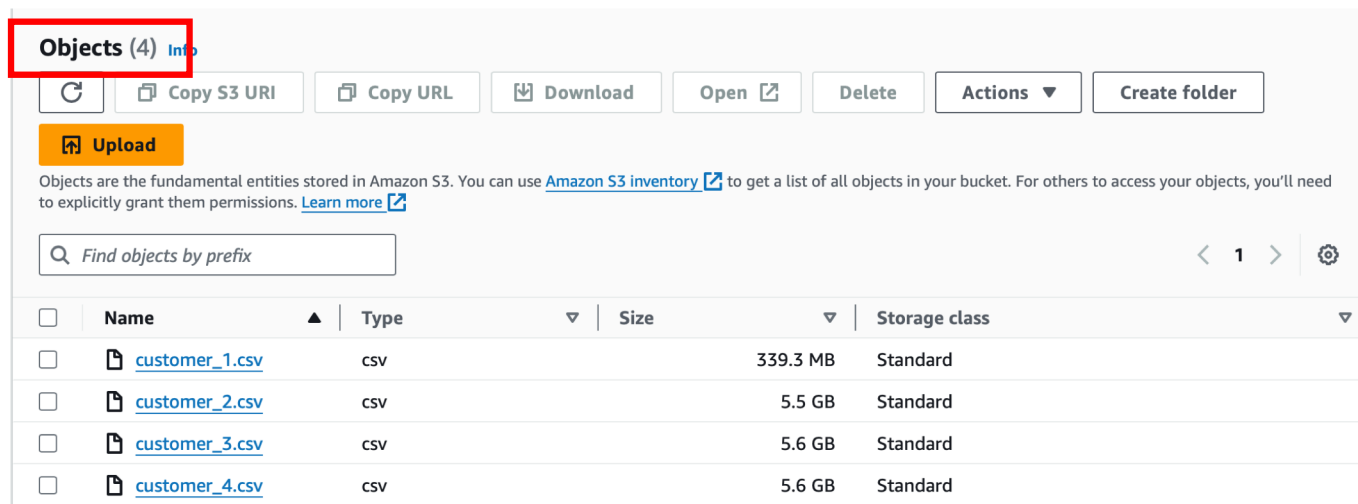
Amazon Simple
Storage Service
(S3)

Google Cloud
Storage

Azure Data
Lake

Data Lake – storing data

- File as a whole (csv, parquet, xls, pdf, jpeg, mp4, mp3)
- No record processing
- Objects



The screenshot shows the Amazon S3 console interface for a bucket. At the top, the 'Objects (4)' tab is selected and highlighted with a red box. Below the tab are several action buttons: 'Refresh', 'Copy S3 URI', 'Copy URL', 'Download', 'Open', 'Delete', 'Actions', and 'Create folder'. An 'Upload' button is also visible. A text block explains that objects are fundamental entities in Amazon S3 and provides links to 'Amazon S3 inventory' and 'Learn more'. Below this is a search bar with the placeholder text 'Find objects by prefix'. At the bottom, a table lists the four objects in the bucket.

☐	Name ▲	Type ▼	Size ▼	Storage class ▼
☐	 customer_1.csv	csv	339.3 MB	Standard
☐	 customer_2.csv	csv	5.5 GB	Standard
☐	 customer_3.csv	csv	5.6 GB	Standard
☐	 customer_4.csv	csv	5.6 GB	Standard

Data Lakehouse

- Data Lake + Data Warehouse
- Combines the capabilities of data lake and data warehouse
- Modify individual records in a Data Lake
- Technology enablers – open table format
 - Iceberg
 - HUDI
 - Delta Lake

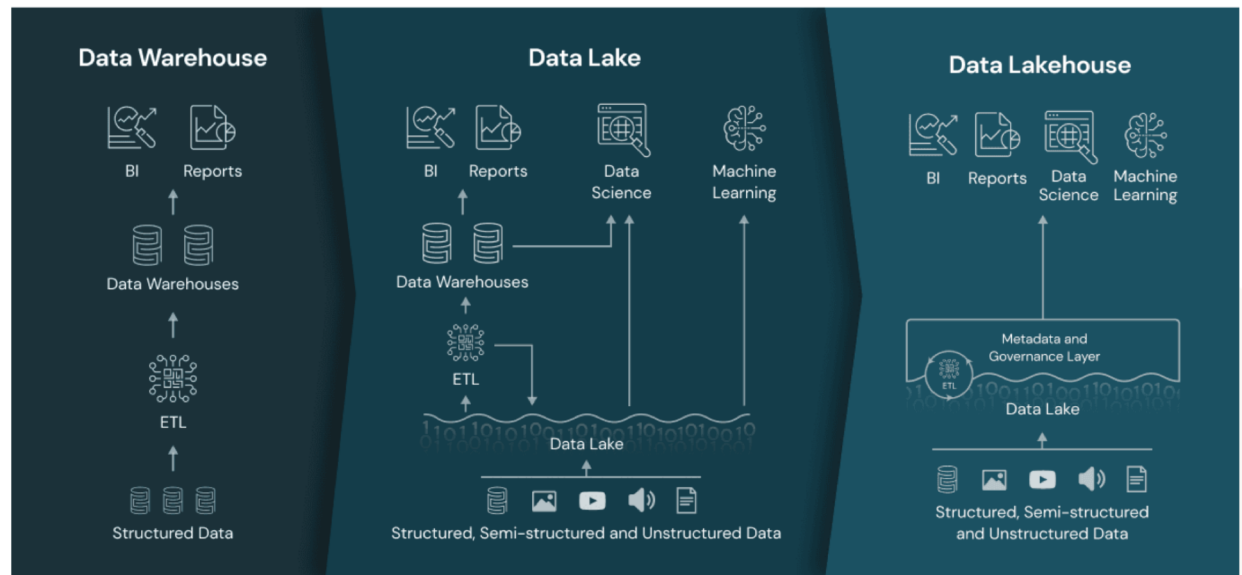
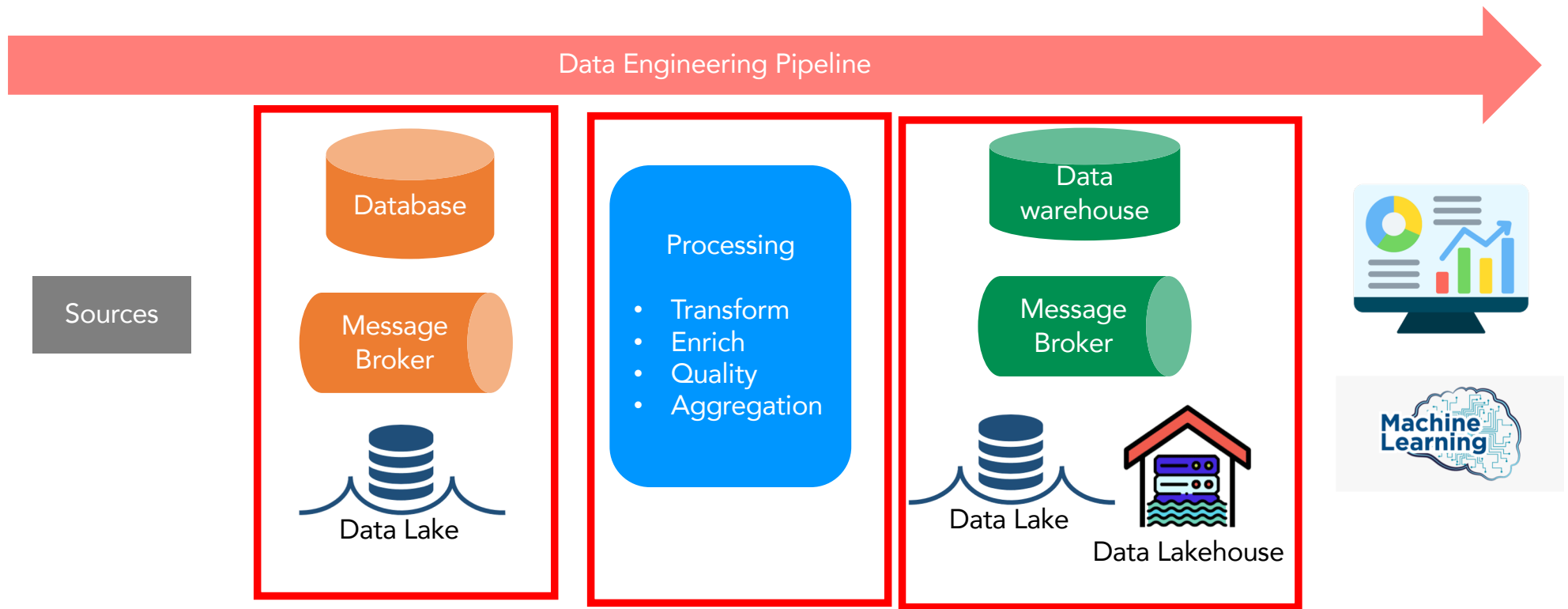


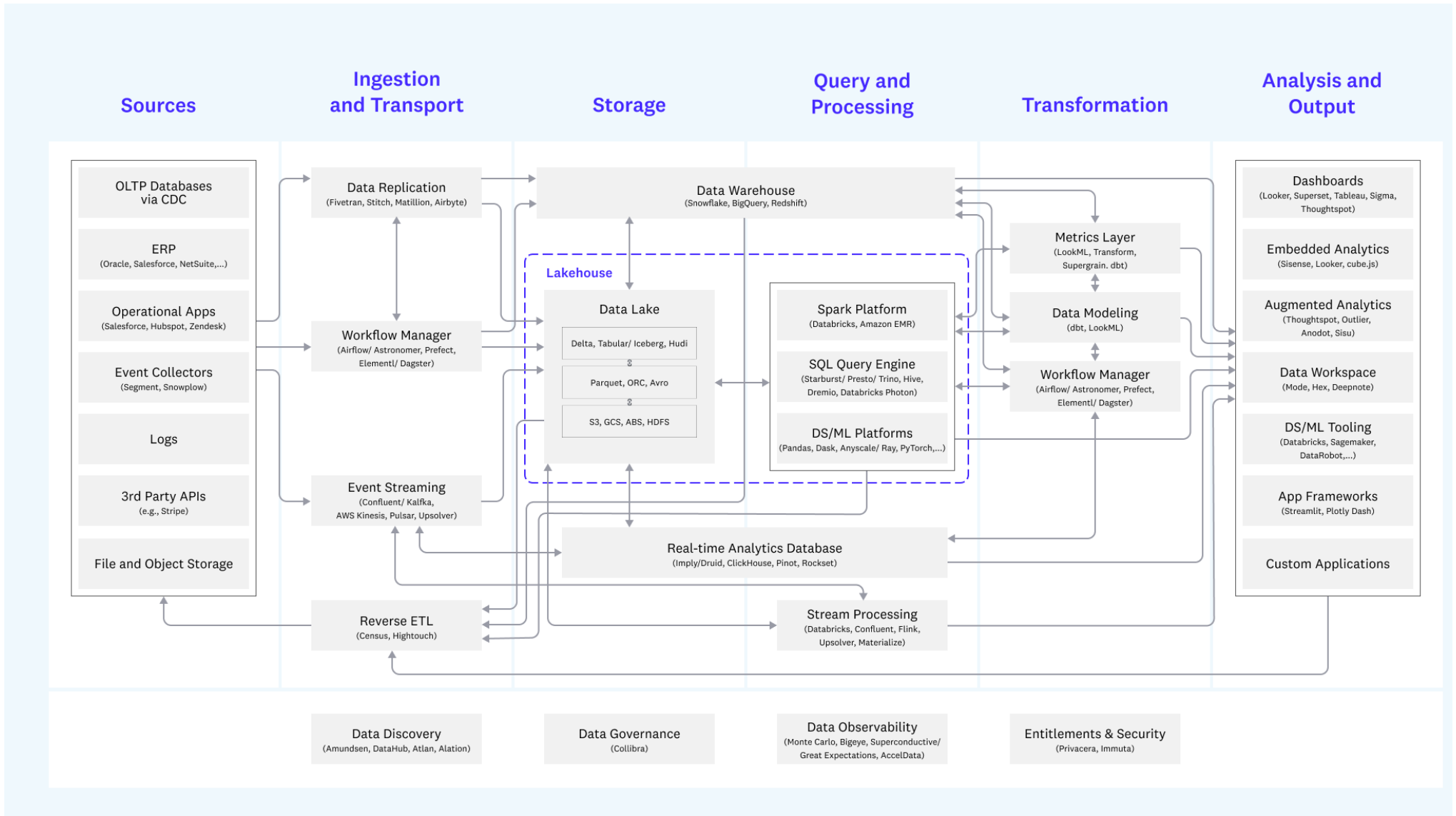
Image courtesy: databricks.com

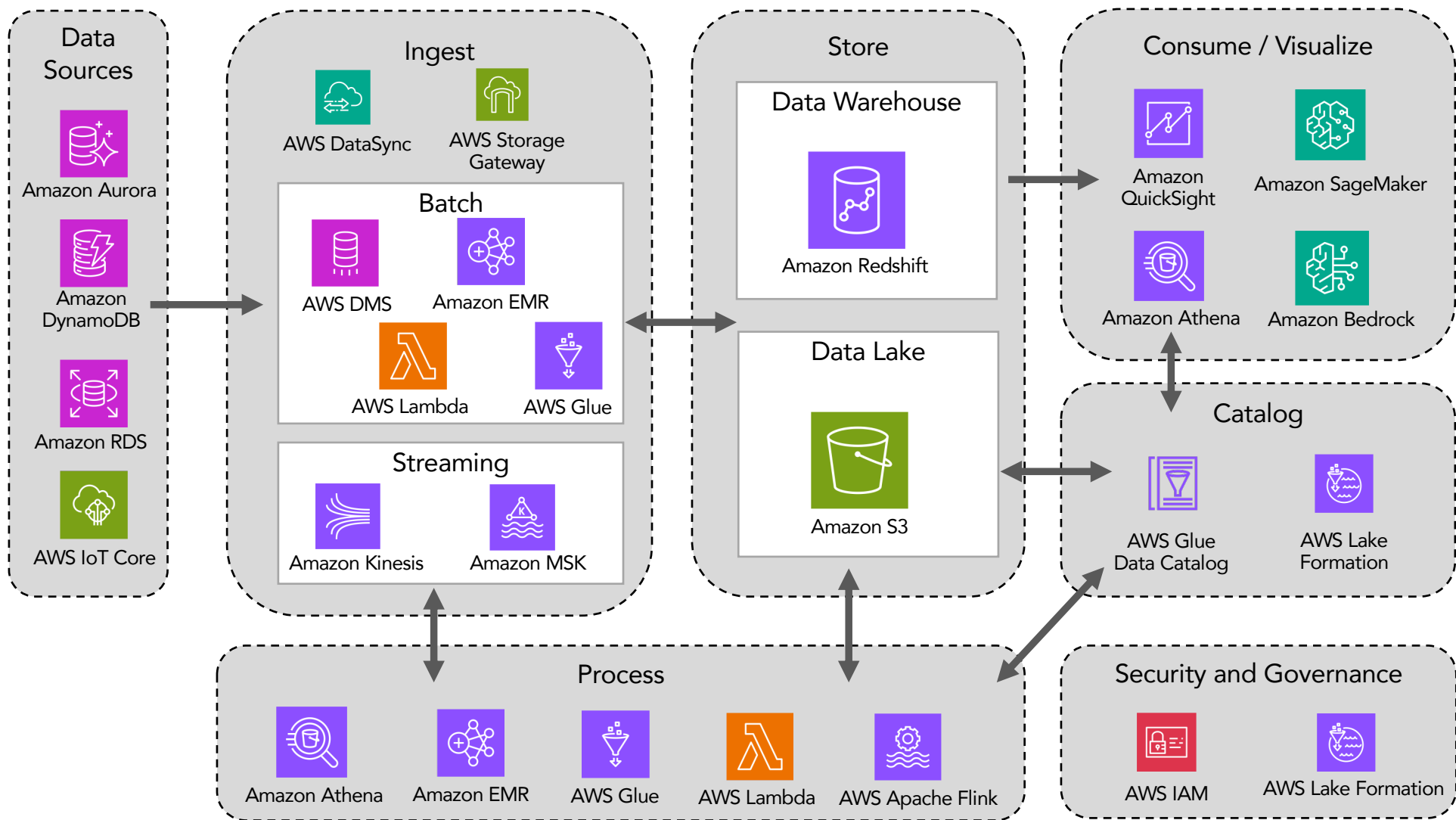
Role of Data Engineer



Use Cases of Data Warehouse (DWH)

- Reporting (tactical and strategic)
- Decision support system (DSS)
- Prediction and forecasting
- Understand business health – historical analysis
- Get crucial business indicators over time with data analytics & reporting
- Data integration - enhance the quality of data by extracting accurate & reliable data from different sources
- Compliance, regulation, auditing
- Personalized customer service
- Fraud analysis
- Input for data pre-processing needed in machine learning





Summary

- How data is generated
- Lifecycle – Data engineering pipeline
- OLTP databases / Operational databases – Relational (SQL) and NoSQL
- OLAP databases / Data Warehouse
- OLTP to OLAP
- Data Mart & Data Mesh
- Data Lake and Lakehouse